

Forking Codex to talk to any endpoint

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```
>_ Fraktal (v0.1.1)
model:    deepseek/deepseek-v4-flash-0731 none  /mode...
directory: ~\dev\fraktal-codex

Tip: New Build faster with Codex.

> what do you think about this fork of codex-rs, isn't it nice?

• Let me take a look at this repo before giving you an opinion.
• Running Get-ChildItem -Force | Select-Object Mode, Name
• Running git log --oneline -20 2>&l; Write-Output '--- remotes ---'; git remote -v 2>&l

Would you like to run the following command?

Environment: Local

$ git log --oneline -20 2>&l; Write-Output '--- remotes ---'; git remote -v 2>&l

1. Yes, proceed (y)
2. Yes, and don't ask again for commands that start with 'git log --oneline -20 2>&l; Write-Output '--- remotes ---'; git remote -v 2>&l' (p)
3. No, and tell Codex what to do differently (esc)
```

I created the fork to be able to run local models after OpenAI made it difficult, and also made the setup very easy.

That is the whole post in one sentence. The rest is the code. We took [OpenAI's Codex CLI](#), restored the Chat Completions wire API that upstream removed, and ended up with an agent that talks to a local Ollama server, OpenRouter, Azure OpenAI, DeepInfra, or any other OpenAI-compatible endpoint by changing one line of TOML. Along the way we turned OpenAI's telemetry off, pointed the agent's own telemetry at a collector we run, and put our branding on it.

So all this to have better control over when one works with sensitive customer data.

The [first post in this series](#) argues why a consultancy needs that control. This one shows how we got it, what broke, and what a book about computer hardware from 2000 has to say about which part of the fork actually mattered.

The problem

Upstream Codex used to support two wire protocols: OpenAI's newer **Responses** API and the older **Chat Completions** API that essentially the entire ecosystem implements. In [discussion #7782](#) they dropped Chat Completions and went Responses-only.

That is a defensible decision for a first-party tool. Responses is where the interesting features live: server-side reasoning items, encrypted reasoning, remote compaction,

WebSocket transport. But it has one consequence. **Codex can only speak to OpenAI-shaped endpoints.**

Everything else exposes Chat Completions and nothing else. OpenRouter, DeepInfra, Groq, Together, Ollama, LM Studio, vLLM, Scaleway's serverless inference, most EU sovereign providers. The usual workaround is a translation proxy: run **LiteLLM** locally, have it expose a `/v1/responses` endpoint, and let it translate down to Chat Completions.

That works. We ran it that way for a while.

A LiteLLM proxy would work as well but that would have been more complicated for the adoption of the tool, also it's more code that we don't have control over, and it was at the time not written in Rust (now they are rewriting it though which is nice).

A proxy is a process to start, a port to keep free, a config file to keep in sync, and a thing to explain to every colleague you onboard. For a tool meant to be handed to consultants who just want to get work done, that is a dealbreaker. So we put `wire_api = "chat"` back.

Quickstart

If you just want a working agent against a cheap model, this is the whole thing. On Windows:

```
irm https://raw.githubusercontent.com/Fraktal-Norge-AS/fraktal-codex/fraktal/main/or-setup.ps1 -OutFile or-setup.ps1
.\or-setup.ps1
```

The script asks for an **OpenRouter key** and does the rest: downloads the binary, verifies its SHA-256 against the published checksum, puts it on `PATH`, stores the key as a user environment variable, and writes the config. Then open a **new** terminal, since `PATH` is read at process start, and:

```
cd C:\path\to\project
fraktal                                # interactive TUI
fraktal exec "what does this repo do?" # one-shot, non-interactive
```

Useful flags:

```
.\or-setup.ps1 -DryRun                # print every action, change
nothing
.\or-setup.ps1 -Model "deepseek/deepseek-v4-pro" # a different model
.\or-setup.ps1 -ApiKey "sk-or- ..."          # non-interactive
```

Three things the script does that you should steal if you write your own installer.

It never overwrites an existing config. If `~/.fraktal/config.toml` already exists, pointing at a local model say, it backs the file up, appends only the missing provider block, and writes

the model selection into a separate profile overlay instead of hijacking your default. You opt in per run with `fraktal --profile openrouter`. Re-running it is safe.

It warns about PATH shadowing. We append to `PATH`, so an earlier entry (a `cargo install`ed build, typically) keeps winning and the binary you just downloaded never runs. Silent shadowing is worse than a warning:

```
$resolved = Get-Command fraktal -ErrorAction SilentlyContinue
if ($resolved -and $resolved.Source -and
    ($resolved.Source.TrimEnd('\') -ne $exePath.TrimEnd('\'))) {
    Write-Warn "Another fraktal is earlier on PATH and will be used instead:"
    Write-Warn "  $($resolved.Source)"
}
```

It reads the checksum from a file, not from `.Content`. This one cost me twenty minutes. For a response PowerShell doesn't consider text, `Invoke-WebRequest`'s `.Content` is a `Byte[]`, and `-split` over a `Byte[]` yields the first byte, `53`, ASCII `'5'`, and not the hash. You get a checksum mismatch on a download that was completely fine. Write it to a file and `Get-Content -Raw` it.

The config file

Configuration lives in `~/.fraktal/config.toml` (`FRAKTAL_HOME` overrides the path). Here is the shape, with all four providers we actually use. Comments translated to English; the repo's are in Norwegian.

```
# Default: our own Ollama server (requires VPN / office network)
model_provider = "fraktal-ollama"
model = "qwen3.5:27b"

[model_providers.fraktal-ollama]
name = "Fraktal Ollama (ml-dev-titan)"
base_url = "http://ml-dev-titan.example.internal:11434/v1"
discover_models = true # [fork] list what the server actually has in /model

# OpenRouter, direct – no translation proxy
[model_providers.fraktal-openrouter]
name = "OpenRouter"
base_url = "https://openrouter.ai/api/v1"
wire_api = "chat" # [fork] direct Chat Completions
requires_openai_auth = false
env_key = "OPENROUTER_API_KEY"
discover_models = true

# Azure OpenAI, with Entra ID auth via an external helper
[model_providers.fraktal-azure]
```

```

name = "Fraktal Azure OpenAI"
base_url = "https://example-aoai.openai.azure.com/openai/deployments/gpt-5-
codex"
wire_api = "responses"
requires_openai_auth = false

[model_providers.fraktal-azure.http_headers]
"api-version" = "2026-02-01"

[model_providers.fraktal-azure.auth]
command = "fraktal-auth"
args = ["token", "--scope", "https://cognitiveservices.azure.com/.default"]
refresh_interval_ms = 300000
timeout_ms = 10000

```

Three things to note.

Profiles are separate files, not TOML tables. A profile `x` is an overlay file `~/.fraktal/x.config.toml` layered on top of `config.toml` when you run `fraktal --profile x`. It inherits `[model_providers.*]`, `[features]` and `[mcp_servers]` from the base config, so an overlay is usually two lines:

```

# ~/.fraktal/openrouter.config.toml
model_provider = "fraktal-openrouter"
model = "deepseek/deepseek-v4-flash-0731"

```

Use the dated model slug. `deepseek/deepseek-v4-flash` resolves to the older `0423` snapshot. The `-0731` slug is both newer *and* cheaper, at 0.09/0.18 per million tokens against 0.14/0.28. Check the exact slug and current price on the provider's model page before switching; this is the kind of thing that changes under you.

The Azure auth block is the interesting one. `command` is an external process the agent shells out to every `refresh_interval_ms` to get a bearer token on stdout. That gives you corporate SSO (device-code login against Entra ID, refresh token in the OS keyring, per-user audit trail) without the agent knowing anything about Entra. Any credential source that can print a token to stdout drops in the same way.

Design rules, and which line was worth fighting for

Before the patches, a short detour into why the first one matters more than the other two put together.

Baldwin and Clark's *Design Rules* (2000) is a book about how the computer industry went from integral machines, where every part was designed for every other part, to modular ones. Their example is IBM's System/360 in 1964. IBM wrote down the interfaces between processor, memory and peripherals, the *design rules*, and let the teams behind each

module work independently as long as they obeyed them. The rules were the visible information. Everything inside a module was hidden, and could change without anyone else caring. The result IBM did not intend was a plug-compatible peripherals industry: once the interface was public, other companies could sell disks and tape drives that fitted an IBM machine, and the customer got to choose.

The mechanism transfers to this case almost without adjustment, since a coding agent is software talking to a server over a documented protocol. Chat Completions is the design rule of the inference market. Nobody voted for it. OpenAI published it, everyone else copied it, and now every GPU cloud, aggregator and local runtime speaks it. The agent is one module. The model server is another. As long as both obey the rule, either can be replaced without the other noticing, which is how Ollama, OpenRouter and a European sovereign provider all work behind the same binary.

When upstream dropped Chat Completions, the agent stopped being a module and became part of one vendor's integral design. The whole value of the fork is that a thousand lines put the module boundary back. Three things follow, and they shaped how the fork is built:

1. **Modularity is valuable because it creates options.** Baldwin and Clark's central argument is that every module you can swap is an option you hold on a better module appearing. A cheaper open-weight model or a provider in the right jurisdiction shows up, and you exercise the option with a `base_url` change.
2. **The interface is where to spend your effort, not the features.** Restoring `wire_api = "chat"` gives up every Responses-only feature and was still the right patch, because it restores the rule. Features can be added later. A missing interface cannot be worked around.
3. **Keep your changes in the hidden modules, away from the rules upstream is rewriting.** That is the fork hygiene section below, in one sentence.

The patches that matter

The fork is about twenty commits on top of upstream, one commit per intent, all prefixed `[fraktal]` so the series stays droppable and rebasable. Most are cosmetic: binary rename, telemetry off, update check repointed, CI gated. Three of them do the real work.

1. Restoring `wire_api = "chat"`

This is the one that makes everything else possible. About 1,400 lines across fourteen files:

<code>codex-rs/codex-api/src/endpoint/chat.rs</code>	98 ++
<code>codex-rs/codex-api/src/requests/chat.rs</code>	496 ++

```

codex-rs/codex-api/src/sse/chat.rs          | 582 ++
codex-rs/core/src/client.rs                 | 123 ++
codex-rs/model-provider-info/src/lib.rs     | 22 +-
codex-rs/tools/src/tool_spec.rs             | 38 ++
...

```

The work breaks down as:

- `WireApi::Chat` back in `codex-model-provider-info`, including deserialisation of `wire_api = "chat"` so the TOML above parses.
- `ChatRequestBuilder`, the chat SSE parser, and `ChatClient` reinstated in `codex-api`, adapted to the current `EndpointSession` architecture, since the transport layer had been reorganised since the removal.
- `create_tools_json_for_chat_completions_api` back in `codex-tools`. Chat Completions and Responses describe tools differently; the agent needs both encoders.
- A `WireApi::Chat` arm in core's client (`stream_chat_completions`) mirroring the Responses path.

Two lessons from doing this as a restoration.

Don't resurrect the old code; port the old behaviour. The temptation is to `git revert` the removal commit. It won't compile, and if you force it into compiling you get an implementation wired to an architecture that no longer exists. The transport now comes from `build_api_transport`, errors go through `provider.map_api_error`, and `map_response_stream` / `handle_unauthorized` take the provider. Making the chat path *mirror* the Responses path structurally is what keeps the patch rebasable. When upstream refactors the transport again, both arms move together.

Some upstream types can't cross the protocol boundary at all. During a rebase, upstream added `InputAudio` to `ContentItem` and `FunctionCallOutputContentItem`. Chat Completions has no representation for it. We drop it as encrypted content on the chat path and document that. A lossy mapping would silently corrupt a conversation.

There is a real trade-off here: **the chat path has no Responses-native features.** No server-side reasoning items, no encrypted reasoning, no remote compaction, no `output_schema`, no WebSocket transport. These are limits of the protocol, so you hit the same wall with any Chat Completions endpoint, whatever model is behind it. For models like GLM, Kimi or DeepSeek V4 Flash driving a normal agentic loop, it is fine. If you need encrypted reasoning, you need a Responses endpoint, full stop.

2. `discover_models`

Codex ships a bundled catalog of OpenAI model metadata. Point it at an Ollama server and `/model` shows you a list of GPT models the endpoint cannot serve, and not the models it can.

The patch adds a `discover_models` flag to `ModelProviderInfo`. When set, the provider's model list is fetched live from its OpenAI-style `/models` endpoint and every reported slug shows up in the picker:

```
[model_providers.fraktal-ollama]
base_url = "http://ml-dev-titan.example.internal:11434/v1"
discover_models = true
```

Push a new model to the Ollama server and it appears in `/model` without anyone editing config. Set the same flag on OpenRouter and pressing `/model` shows you their whole catalog, which is a really nice way to try a new release the day it lands.

Two design decisions worth copying:

Discovered models replace the bundled catalog. They do not merge with it. For a provider that reports its own models, that report is authoritative, and merging just repollutes the picker with models the endpoint can't serve.

Discovered models get generic default metadata, not "fallback" metadata. OpenAI-style `/models` only reports slugs, so context window, pricing and capabilities are unknown. There was an existing `model_info_from_slug` path that guesses and flags the result as a fallback, but here the absence of metadata is expected, and not a failure, so a separate `discovered_model_info` keeps the picker clean of spurious warnings.

The flag is local-only and deliberately not plumbed through the remote thread-config surface.

3. Making MCP tools work with local models

This is the patch I'd most like to see upstream, because it's a host-side bug that only a host can fix.

Codex exposes **MCP** tools to the model as `mcp__<server>__<tool>`, for example `mcp_wren_charts__list_models`. The prefix, the namespace (which is your config key) and the `__` separator are all added host-side; the MCP server itself only ever offers flat names like `list_models`.

Upstream sets `ProviderCapabilities::namespace_tools` to true for every provider, so it emits Responses-API `type: "namespace"` tools unconditionally. Only the OpenAI GPT-5 family implements that convention. Every other model sees the namespace as an opaque wrapper and cannot call the tools inside it. **MCP tools are effectively invisible to every non-OpenAI model.**

Flattening namespaced tools into plain top-level functions fixes half of it. The other half is that local models don't reproduce the wire name exactly:

- the name comes back **flat**, with no separate namespace field, and
- weaker models substitute the `__` separator with `.` or `:`. `qwen3.5:27b` did exactly this.

The result was that upstream's exact-match lookup returned `unsupported call` for every local model, including `gemma4:26b`, which emitted the perfectly correct `mcp_wren_charts_list_models`, because the flat name was never split back into namespace and tool.

The fix is a tolerant fallback in `core/src/tools/registry.rs` (`resolve_fuzzy_mcp_name / canonical_tool_key`). If the exact lookup misses, normalise the namespace/name boundary by collapsing runs of `.: - _` to a single `_`, then resolve to the one registered MCP tool that matches. **Ambiguous matches are rejected**, so it never guesses, and that constraint is what keeps the fallback safe. There are unit tests for it in `registry_tests.rs`.

With that, both `qwen3.5:27b` and `gemma4:26b` call MCP tools end to end.

One caveat if you go down this road: MCP Apps servers can expose interactive `ui://` resources ([SEP-1865](#)). A terminal doesn't render iframes, so in a CLI you get the data plus a saved PNG path instead of the interactive UI. Tools without a UI render as text as you'd expect.

Fork hygiene

Codex moved 1,288 commits between two of our rebases. If you are going to maintain a fork of an upstream moving that fast, the branch model matters more than the patches.

The forking does work but it's a lot of changes in upstream so it's hard to keep up, so it's good to keep the commits to a minimum and have focused changes that are unlikely to be in a high churn code path.

Here is what that looks like in practice.

`main` mirrors upstream untouched, and all fork work lives on `fraktal/main`. In our case that arrangement was forced by an org policy that locks every repo's default branch to `main`, which is annoying, since people landing on the GitHub home page see upstream's README. It turns out to be the right model anyway. A pristine mirror makes `git log upstream/main..fraktal/main` the live, exact list of everything you own.

One commit per intent. Not one commit per file, and not one big "fork changes" commit. Each patch is independently droppable and independently rebasable. When upstream reorganises a module you touched, and they split `codex-mcp/src/connection_manager.rs` into a directory on us, you fix one commit, not a merge blob.

Gate upstream CI instead of deleting it. We initially deleted `.github/workflows/bazel.yml`, which required remote-cache secrets we don't have. Every single time upstream touched that file we got a delete/modify conflict. Gating it instead:

```
if: github.repository == 'openai/codex'
```


That self-skips on the fork, keeps the scaffolding intact for when we wire up our own credentials, and never conflicts again. Same trick for the release pipeline.

Separate your state directory. Settings, auth, history, sessions and logs live under `~/.fraktal`, not `~/.codex`, so the fork and a stock Codex install can coexist on the same machine without corrupting each other's state. `CODEX_HOME` is still accepted as a fallback so upstream tooling and scripts keep working unchanged. The repo-local `.codex/` project directory is deliberately left alone, since the fork should still read project config a team has already committed.

Write the rebase runbook before you need it. Ours records what broke on each sync and why. With it a rebase is a scheduled hour. Without it a rebase is an afternoon of archaeology.

Where this leads

Once the agent can talk to any OpenAI-compatible endpoint, moving between them is a `base_url` change:

```
[model_providers.eu-sovereign]
name = "EU serverless inference"
base_url = "https://api.example-eu-provider.com/v1"
wire_api = "chat"
env_key = "EU_PROVIDER_API_KEY"
```

You keep the same binary, the same agent loop, the same MCP tools and the same muscle memory, and you change the jurisdiction, the processor agreement and the price. That property is what the whole exercise was for, and the [business post](#) covers why it is worth more to a consultancy than any individual model choice.

It is also why that last config block exists at all. The provider behind it did not, eighteen months ago. Call the trend reshoring, digital sovereignty, or just not wanting your customers' data under a jurisdiction that can subpoena it: in an unstable world, more of the people we work for want to know which country the inference happens in, and a growing number of European providers now sell exactly that. None of them speak Responses. All of them speak Chat Completions. The design rule did its job.

One of the cosmetic patches turns the agent's own telemetry off by default. That is the right default for a tool handed to consultants. Pointed at a collector you run yourself it becomes useful again, and the [next post](#) is about getting real cost numbers out of it, including the part where Codex's token metrics report zero.

If you fork a fast-moving upstream yourself: keep the patches few, keep them off the paths upstream is rewriting, and write down what broke on every rebase. The fork is a lightly patched copy of someone else's very good tool. Most of what it bought us came from putting one interface back.